

Volcanoes to vugs: Demonstrating a FAIR geochemistry framework with a diverse application of major and trace element data through the AuScope EarthBank platform

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ABSTRACT

Major, minor and trace geochemistry is perhaps the most commonly collected form of analytical data within the Earth sciences, either as the sole approach to a research question or to supplement other methods such as geochronology, thermochronology or isotopic studies. With increasing access to instrumentation and diversification of research applications, inorganic geochemistry data is being produced in volumes which exceed the capacity of existing data management systems, reducing the ability of data to be found, accessed and reused as part of the data lifecycle, thus consigning much data as 'single use'. Lack of visibility additionally removes the ability to critically assess data quality and validity of interpretations, which is vital for maintaining a healthy research ecosystem. Here we propose a new, flexible inorganic geochemistry reporting schema available within the open access EarthBank platform, as a service to store, interpret, disseminate and explore geochemistry data. The application of this data structure, augmented by purpose-built data interrogation tools, is demonstrated using real-world case studies, including volcanic evolution and magma genesis, mineralisation prospectivity, and catchment provenance and contamination through stream sediment geochemistry. The combination of data schema and integrated data analytics through EarthBank offers a service beyond a basic repository, and allows for active exploration and interpretation of in-house and community geochemistry data, providing a resource beneficial across the entire data lifespan, from generation to publication and reuse.

1. Introduction

Geochemistry is a broad and highly diverse discipline of study within geosciences, with applications to a wide variety of research areas required for the understanding of our planet and the processes that govern it. Geochemical abundance data, that is the chemical composition of rock, mineral, soil, or other material of interest, provides vital

information that enhances our knowledge of the composition, structure and evolution of the Earth and the formation and preservation of critical geological resources (e.g. Pearce and Cann, 1973; McLennan et al., 1993; Taylor and McLennan, 1995; Li and Agee, 1996; Hofmann, 1997; McLennan, 2001; Dilek and Furnes, 2011; Henderson, 2013; Keller and Schoene, 2018; Barnes et al., 2021). This discipline also extends beyond the terrestrial 'hard-rock' studies, with applications extending to the

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extraterrestrial (Buseck, 1977; Koeberl, 1990; Korotev, 2005; Barrat et al., 2012; Taylor, 2013; Rampe et al., 2020), marine system (Moore et al., 1983; Boyle and Keigwin, 1985; Maier-Reimer, 1993; Druffel, 1997; Ikeda et al., 2018), modern landscape evolution and provenance (Carranza, 2004; Singh, 2010; Armstrong-Altrin et al., 2015; Kirkwood et al., 2016; Amorosi et al., 2022; Wang et al., 2022), waste remediation and reprocessing (Jamieson, 2011; Jackson and Parbhakar-Fox, 2016; Parbhakar-Fox, 2017; Vriens et al., 2020), and environmental monitoring (Lettoof et al., 2021; Naciri et al., 2023) fields of research. Furthermore, in many cases geochemical concentration data also acts as a complementary tool to other data types and disciplines including geo/thermochronology, isotopic signatures, mineralogy, and petrology to refine or enhance these resources.

Today the geochemical community enjoys unprecedented access to analytical equipment such as inductively coupled plasma mass spectrometry (ICP-MS), x-ray fluorescence (XRF), electron probe microanalysis (EPMA), thermal ionisation mass spectrometry (TIMS), and inductively coupled plasma optical emission spectroscopy (ICP-OES) systems to highlight a few. These techniques are able to acquire quantitative geochemical data for major, minor and trace elements from a range of analyte materials, including thin-sections, polished sections, grain mounts, crushed powders, pressed powders and bulk material digestions. This growing availability of fast, semi-automated and increasingly cheap and accessible analytical instrumentation has facilitated an explosion in the generation and publication of geochemical data from university, government, and commercial laboratories, which are disseminated via a variety of different outlets, such as academic publication, public or internal reports, or institutional repositories (Klöcking et al., 2023). Increasing volumes of geochemical data from a broad array of sources, instrumentation and processing software have, however, seen data acquisition outpace systems for dissemination, synthesis, evaluation and interpretation of geochemical information. This decoupling between data collection and data management has brought into focus the need for improved community and digital resources and behaviours to maximise the value and openness of geochemistry resources, as the current data landscape has entrenched a compartmentalised environment of data and metadata reporting protocols, terminologies and analytical practices between different laboratories and institutions both nationally and globally. Such variations inhibit the ability for the wider research community to access, evaluate and reuse geochemical data produced by other parties, despite this data often being acquired through projects supported by public funding and subject to open data requirements.

In response to these concerns, the AuScope Geochemistry Network (AGN) initiated a project in 2019 to reimagine the model for capturing and preserving Earth science data, by promoting data and metadata upload directly from laboratories and scientists as a part of the research process (Boone et al., 2022). This approach seeks to collect instrument metadata and processed results on a bespoke geochemical platform prior to interpretation and publication of results, to capture a greater proportion of produced geochemical data and allow researchers to statistically interrogate and integrate their results with other publicly available data available through the platform. Originally launched under the name AusGeochem (Boone et al., 2022), the platform was rebranded EarthBank in 2025 to reflect the growing international availability of this service and improvements in Open Science practices, to now allow users view published datasets without requiring login or registration.

This endeavour has required the creation of new (meta)data standards and reporting templates needed to store information in a standardised format, allowing for direct comparison, sharing and interoperation of data from different sources (Boone et al., 2022; Boone et al., 2023). Data stored within this digital architecture is designed to adhere to the principles of data that is findable, accessible, interoperable and reusable (FAIR; Wilkinson et al., 2016), to promote reuse and data transparency in accordance with globally defined principles. Here we

detail the AGN reporting model for inorganic geochemistry abundance data, which has been designed as a flexible resource to accommodate the breadth of studies and applications commonly observed within this discipline, ranging from nano-scale to bulk sample analyses of materials including rocks, minerals, soils or fluids. This reporting schema provides not only a template for geochemistry data upload and preservation, but allows data within the EarthBank platform to be statistically queried, spatially plotted or visualised through flexible plotting options with user and community data in order to maximise the value of new and legacy data and improve research outcomes.

2. Methods

Inorganic geochemistry is a broad field which may not be easily compressed to a single format, but is defined through the collection of quantitative elemental and/or inorganic chemical species concentrations for a given material. Data collected varies by research topics and analytical instrumentation, but may include elements and isotopes from a breadth of the periodic table and combinations thereof as chemical compounds or homonuclear species. More so than perhaps any isotopic dating system or geoscience method, geochemical abundance data is acquired using a wide array of analytical techniques applied at different scales from nano-scale to whole-rock or even surface imaging, across a variety of analyte materials. Such complexity requires flexible data and metadata reporting standards and vocabularies, which has previously inhibited widespread adoption of standards required for open data and interoperability. While a number of standards exist for geochemical abundance data, the current resources are largely fragmented and often limited specific methodologies, sample types or use cases. In order to satisfactorily store and access geochemistry abundance data while balancing the need for usability to the wider Earth science community, a new data schema has been established by the AGN which allows interoperability with other isotopic and geochronologic systems, as is commonly required for maximising the value of these data types. The new reporting framework was developed in collaboration with an expert working group of experienced international geoscientists and technicians in order to satisfy a large suite of applications and laboratory practices. This template is available through the EarthBank platform, and machine-actionable vocabularies are available through Australian Research Data Commons (ARDC) Research Vocabularies Australia (RVA) service at <https://vocabs.ardc.edu.au/viewById/689>, and associated vocabularies.

Geochemical abundance data and associated metadata in the AGN model are stored in a custom simple knowledge organisation system (SKOS) format of four data tables within a relational database framework (Fig. 1), which preserves data relationships and allows data to be easily queried, displayed, analysed and exported. The first of these tables is the *Geochemical Data Point* table, which stores metadata on the geochemical analysis such as the sample name, analysis type, mineral and analysis scale (Table S1). The *Geochemical Data Point* table is linked to other tables containing metadata on the sample itself, including co-ordinates of sampling location, lithology type and stratigraphic section (full outline of relational framework and sample metadata schema in Boone et al., 2022). The analytical technique used to collect data may be specified within this table, and geochemical data can additionally be linked to an analytical session through a common *analyticalID* identifier (or 'string'), connecting samples and reference materials analysed in a single session. This allows reference material results to be stored and viewed in the same way as results for unknowns, in order to assist with assessment of data quality. In the future, this will also provide links directly to analytical metadata for a given session in order to provide further insights to data quality.

Connected to the *Geochemical Data Point* table are three child tables which hold the reported geochemical data values and additional metadata on discrete analyses within the sample. The *GC Aliquot* table stores details of individual analyses of a sample, such as multiple aliquots or

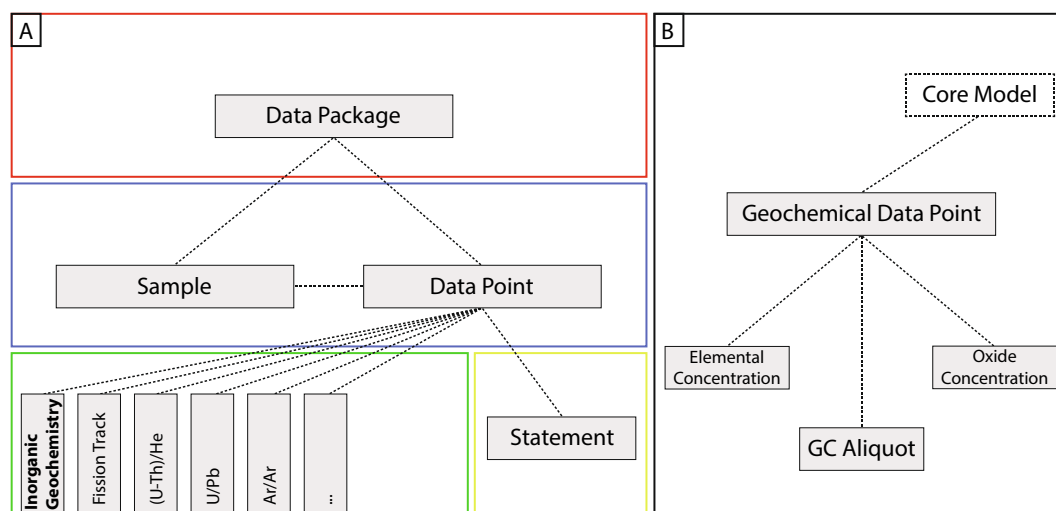


Fig. 1. Entity relationship diagram of the EarthBank platform. (A) Relationship model for samples and data points within EarthBank. Metadata describing samples (see Boone et al., 2022) may be linked to multiple data types, including inorganic geochemistry, fission track, (U–Th)/He (Boone et al., 2023), U/Pb and Ar/Ar. This approach allows for multiple data types to be associated with the same sample (Boone et al., 2022). Sample and geochemical data may be stored within a Data Package for data relating to a discrete project or study region, and at this level data can be shared with collaborators, made public for the community to use and reference, and a DOI may be generated for the Data Package to facilitate appropriate citation and author acknowledgement. (B) the AGN inorganic geochemistry model, which consists of the *Geochemical Data Point* to store procedural metadata, and child tables of *GC Aliquot*, *Elemental Concentration* and *Oxide Concentration* to store analytical results and context of individual analyses.

laser ablation spots on grains or materials related to the sample (Table S2). The *GC Aliquot* table additionally contains a field to store the total wt% of analysed chemical species, which is often reported from analytical techniques such as EPMA and XRF for data quality screening. Quantitative geochemical abundance data and uncertainties are stored as numerical values within the *Elemental Concentration* (Table S3) and *Oxide Concentration* (Table S4) tables, linked to the analyses described in the *GC Aliquot* table. All elemental values within the *Elemental Concentration* table are stored with units of $\mu\text{g/g}$ (equivalent to ppm), while all oxide values within the *Oxide Concentration* table are stored with units of wt% (equivalent to % m/m) to ensure internal consistency and allow direct comparison with other data. The *Elemental Concentration* table additionally records the measured isotopic mass used for determination of elemental abundance within an analysis (where required for mass spectrometry methods), with measured mass stored as an integer with units of amu.

In practical terms and for data upload, Tables S2–S4 can be considered as a single table which conforms to conventional geochemical abundance data conventions. This merged table is outlined in Table S5 and allows front-end upload and download to be achieved via a two-table format of the *Geochemical Data Point* table and merged *GC Aliquot*, *Elemental Concentration* and *Oxide Concentration* table (referred to as the GC Concentrations table). Published data can be subsequently downloaded by a user, allowing export of complete data tables or as shapefiles for use in external software if required.

As geochemical abundance data is often ancillary to other geoscience approaches such as geochronology, thermochronology or isotopic studies, the relational design additionally allows multiple data types to be connected via a shared sample of interest (Fig. 1A; Boone et al., 2022; Boone et al., 2023). This provides users to investigate more complex geological questions which require multiple data types, such as considering the effect of mineral chemical composition on the preserved thermochronological age in order to predict an appropriate thermal evolution model (e.g. Boone et al., 2023; McMillan et al., 2024), or triaging the geochronological age of single detrital grains with grain chemistry to discriminate potential populations and provenance terranes (e.g. Nixon et al., 2024). Multiple data type studies and data resources have already been published within the EarthBank platform, combining fission track, (U–Th–Sm)/He, low-temperature thermal

history and U/Pb data with inorganic geochemistry abundance data to improve research outcomes (McMillan et al., 2024; Nixon et al., 2024). The relational design of EarthBank allows for flexible combinations of data pairings (Fig. 1A), as well as linking between new data models which may be added in the future. In this way, EarthBank and the underpinning data schemas may be applied not merely for inorganic geochemistry, but for multi-faceted studies to address a complex and diverse geoscience world.

3. Results

The following case studies represent a selection of geochemical abundance data from a number of regions, scales, and research questions to demonstrate the application of relational geochemical data visualised within a bespoke, spatially oriented platform. Using in-built data visualisation and analysis tools these, and other, datasets can be studied and interpreted within a spatial and geological context to aid in the research pathway. As the data is open access, additional analysis via functions and software which may not be available through the EarthBank platform can be conducted by researchers by downloading the relevant datasets to work with the data offline at their convenience, or connecting online via API to pull data to preferred software or analysis tools. Further details on API access can be found on the EarthBank website at <https://www.auscope.org.au/earthbank-help#API>.

3.1. Major and trace elements – Sunda-Banda Arc

Geochemistry of whole-rock or key mineral phases has long been applied to igneous, and particularly volcanic, lithologies in order to inform on parental melt compositions, such as the mantle or underlying continental or oceanic crust, and study magma evolution processes including fractionation, country rock assimilation, oxidation, and magma ascent and eruption styles (e.g. O'Hara and Mathews, 1981; Le Bas et al., 1986; Middlemost, 1994; Kemp et al., 2007; Kelley and Cottrell, 2009; Richards, 2015). Through studying the geochemical characteristics of modern systems and isolating sensitive elements and element combinations and their response across key igneous processes, diagnostic geochemical signatures can be applied for samples from extinct systems in deep geological time to inform the evolution and

processes in the ancient Earth.

Here we present a detailed volcanic geochemistry dataset from one such modern volcanic system. Whole-rock and key mineral geochemistry from the Sunda-Banda Arc has been collected and studied across numerous decades by Prof. John Foden and collaborators (Foden and Varne, 1980; Foden, 1983; Ramsay et al., 1984; Wheller et al., 1987; Stolz et al., 1988; Stolz et al., 1990; Elburg and Foden, 1998; Turner and Foden, 2001; Elburg et al., 2002; Elburg et al., 2003; Turner et al., 2003), which has been donated and compiled in EarthBank and is now openly available through this platform (Fig. 2). This dataset contains geochemical data for 473 volcanic samples across the Sunda-Banda Arc, including whole-rock digests and individual mineral phase analyses, and is openly available at <https://doi.org/10.58024/AGUAEBE2CBB4> (Nixon and Foden, 2024). The collective geochemistry resource illustrates characteristic geochemical signatures of arc regimes, which may be visualised and explored using tools within the EarthBank platform.

The Sunda-Banda Arc is a complex expression of island arcs,

ophiolitic accretion and remnant continental material, situated at the junction between the Pacific, Asian and Indo-Australian plates (e.g. Hamilton, 1979; Harris, 2006; Nugroho et al., 2009; Harris, 2011). The extensive volcanic system comprises the Sunda Arc, which extends ~6000 km from Myanmar to the Indonesian island of Flores, and the Banda Arc, which extends a further ~2300 km from Flores to the Indonesian Maluku Islands (Fig. 2A). Activity within the Sunda-Banda Arc has been long lived, with subduction related volcanism extending from the Cretaceous (Lytwyne et al., 2001; Harris, 2006) to present day, and thus it provides an excellent natural laboratory of collisional volcanism and subduction related melt evolution.

Using the Foden Sunda-Banda Arc dataset within the EarthBank platform we provide a case study for exploring igneous geochemical data across a number of volcanic centres across the arc, beginning with the island of Lombok (Fig. 2B). Geological samples from the Rindjani Volcano on Lombok shows a calcalkaline suite of lavas, including ankaramite, basalt, andesite and dacite, where lithologies are as

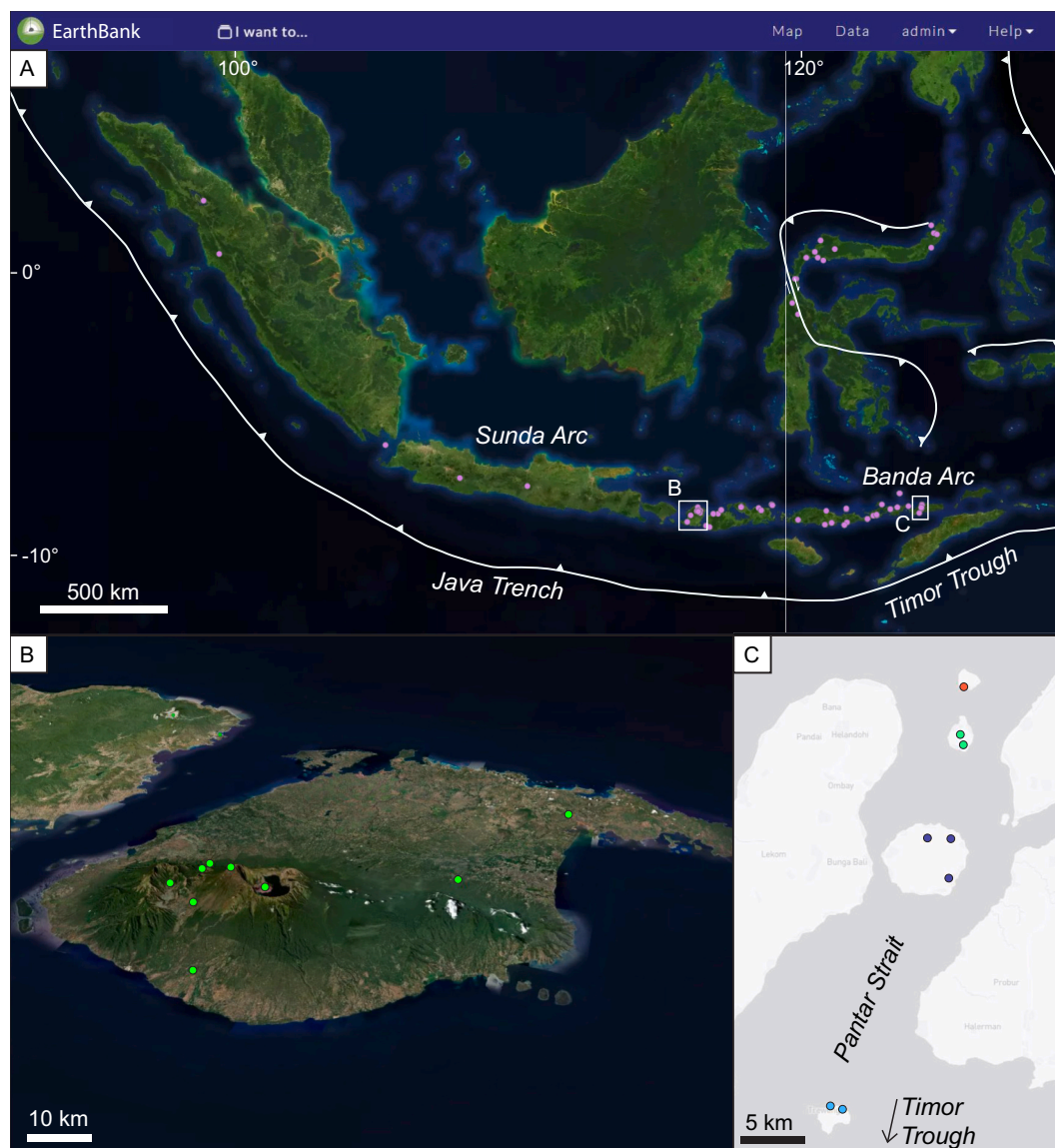


Fig. 2. The Foden Sunda-Banda Arc compilation viewed within the EarthBank platform. (A) Satellite map of the Sunda-Banda Arc region, with sample locations shown as viewed in EarthBank. Subduction zones have been annotated over the platform view to provide geological context. (B) Locations of volcanic samples on the island of Lombok, as seen on a 3D satellite view within EarthBank. (C) Volcanic samples in the Pantar Strait viewed over the 'light' map available within EarthBank. Samples are coloured by groups selected using the multi-select tool, defined by the island of collection. Islands from north to south are as follows: Kisu (red), Ternate (green), Pura Besar (dark blue) and Treweg (light blue). (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

provided within the reporting literature (Foden, 1983). These lithologies can be largely validated using a total alkali silica (TAS) plot generated directly within EarthBank (Fig. 3), to provide an additional check on sample lithologies assigned at collection.

Geochemistry of the evolving volcanic suite can be further investigated by plotting on Harker diagrams (Fig. 3), which may assist in assessment of the potential melt source and evolution. Basaltic lavas from the Rindjani Volcano exhibit higher CaO and MgO contents than would be typically derived from a tholeiitic source of the subducting oceanic slab (Foden, 1983), which suggests mantle wedge or crustal contribution as the major potential sources. In the Lombok region, a significant crustal contribution is unlikely as the underlying crust is quite thin (Métrich et al., 2018; Ariyanto et al., 2020), and the extensive mafic volcanics associated with the Rindjani Volcano are inconsistent with crustal melting at arc geothermal gradients. This suite is, therefore, more consistent with a mantle wedge derived melt that has undergone fractional crystallisation to erupt a mafic to felsic calcalkaline sequence (Foden, 1983; Métrich et al., 2018).

The Pantar Strait hosts a linear array of extinct volcanic islands on oceanic crust approximately orthogonal to the Timor Trough in the Banda Arc (Fig. 2C). From south to north, the volcanic islands of Treweg, Pura Besar, Ternate and Kisu appear at increasing distances from the arc front, and at increasing distance above the northward dipping subducted slab (Fig. 2C). Using samples from the Foden collection, this sequence provides an opportunity to further assess geochemical signatures of each volcanic centre to study the evolving sub-arc processes and melt sources as the slab subducts northwards (e.g. Dickinson, 1975; Woodhead and Johnson, 1993; Elburg et al., 2002).

As the volcanic centres progress northwards, there are notable systematic differences in whole-rock geochemistry of the andesites and basaltic-andesites. Frontal arc volcanoes demonstrate the highest concentrations of Fe_2O_3 , MgO, and CaO, which decrease towards the rear

volcanoes, while concentrations of alkali elements increase towards the rear volcanoes (Fig. 4). Arc transect trends in major element trends are also shown within trace and rare earth patterns, with both trace and rare earth element abundance increasing from the frontal to rear volcanoes (Fig. 4).

Further evidence of fore-arc to rear-arc geochemical change is observed in elemental ratios of Pb/Ce against Ba/Th, both of which decrease away from the frontal volcano (Fig. 4). High Pb/Ce and Ba/Th ratios are indicative of melts which have experienced a strong fluid component sourced from the dehydration of oceanic crust (Turner et al., 1997; Elburg et al., 2002). A complementary geochemical trend is observed in elemental ratios of Th/Ce against U/Pb which exhibit similar positive correlation, with both ratio pairs increasing towards rear-arc volcanic centres (Fig. 4). Together these ratios suggest an increasing contribution of continental material to the melt (e.g. Hawkesworth et al., 1997), sourced either from the subduction of continentally derived sediments or the continental crust itself as the Indo-Australian continental plate subducts below the overriding Asian plate (Elburg et al., 2002).

These volcanic islands, therefore, illustrate a local scale expression of magma evolution from fore-arc to back-arc, where fore-arc magmas are characterised by fluid melting and dehydration of the subducting oceanic crust. Away from the arc front, more distal volcanoes exhibit a stronger continental signal as continental sediments or crust are assimilated to the melt (Elburg et al., 2002). These interpretations represent only a subset of the analysis available across the Sunda-Banda Arc, yet the flexibility of geochemical plotting and visualisation available through the EarthBank platform allows users to further explore this compilation and any additional datasets to realise the full potential held within geochemical data across a range of applications.



Fig. 3. Whole-rock geochemical data from the Rindjani Volcano on the island of Lombok plotted in EarthBank. Plots include a total alkali silica (TAS) discrimination plot (Le Bas et al., 1986) and Harker diagrams. Samples are coloured by lithology assigned at collection (Foden, 1983); ankaramite (orange), basalt and basaltic-andesite (green), andesite (red) and dacite (purple). (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

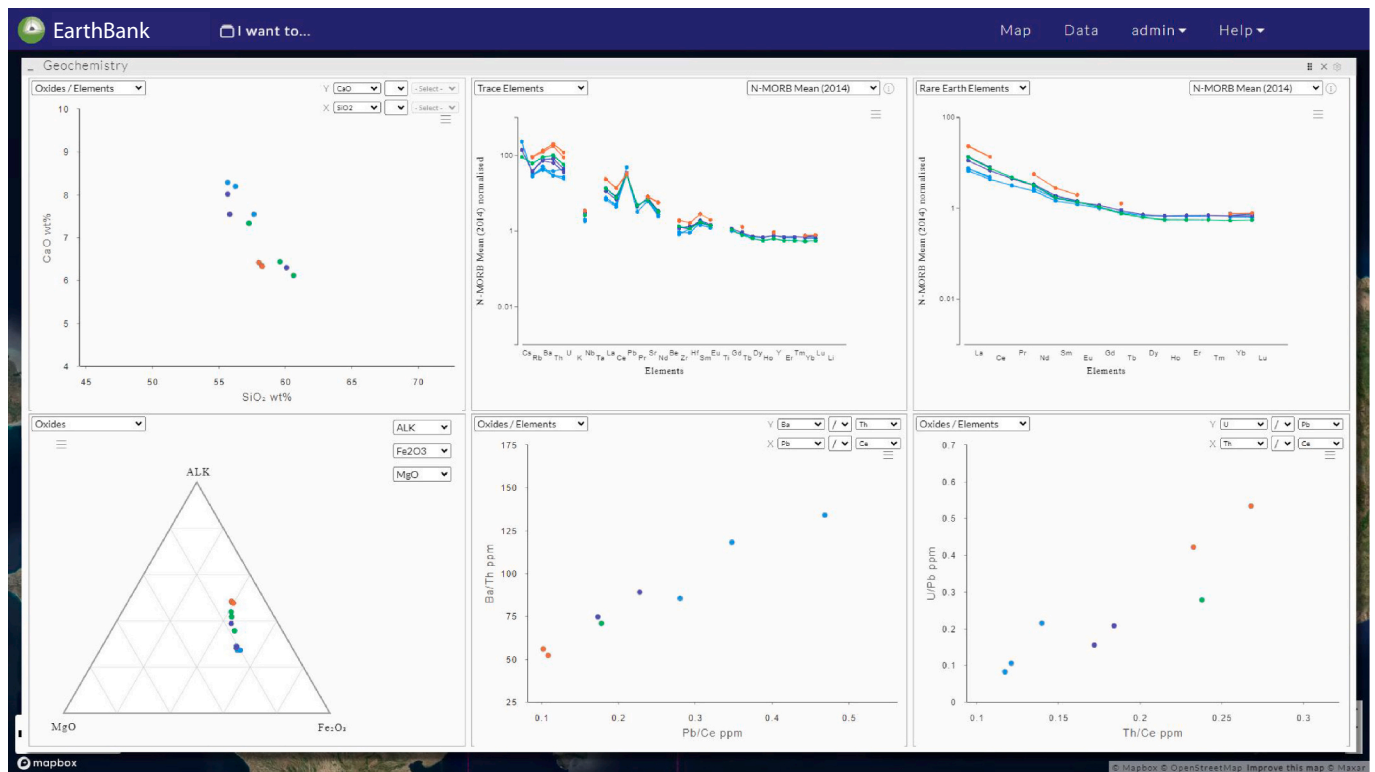


Fig. 4. Whole-rock geochemical plots for volcanic samples from the Pantar Strait, produced within EarthBank. Sample colours correspond to the volcanic islands as shown in Fig. 2B, as follows: Kisu (red), Ternate (green), Pura Besar (dark blue) and Treweg (light blue). Major element plots include CaO vs. SiO₂ wt% bivariate plot and MgO–Fe₂O₃–ALK ternary plot, where ALK = K₂O + Na₂O. Trace element and rare earth element spider plots are normalised to N-MORB values of White and Klein (2014). Trace element ratio plots were created using the custom ratio functionality within EarthBank, and include Ba/Th vs. Pb/Ce and U/Pb vs. Th/Ce plots. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

3.2. Major and trace elements – Ni-Cu-Co-PGE mafic-ultramafic mineralisation

Another common application of igneous geochemistry in Earth science is in the characterisation and exploration of economic mineralisation for a wide array of critical commodities. Most commonly acquired as whole-rock geochemical assays, inorganic geochemistry is routinely used to assess chemical fertility for mineralisation, vector towards mineralised regions, and evaluate the quality of a potentially mineralised area (e.g. Large and McGoldrick, 1998; Holwell and McDonald, 2006; Hao et al., 2017; Wells et al., 2020). In connection with other regional data such as remote sensing and lithological mapping, this can be used to reduce exploration risk and converge on mineralisation targets.

One critical commodity group discovery which can be aided by the collection and availability of geochemistry data is nickel, copper, cobalt and platinum group elements, or Ni-Cu-Co-PGEs. Here we present a world class geochemistry dataset examining komatiite Ni-Cu-Co-PGE mineralisation and regional hosts from Western Australia, produced and compiled by Dr. Steven Barnes of the Commonwealth Scientific Industrial Research Organisation (CSIRO) from the mid-1980s to 2021 (Barnes et al., 2015; Barnes and Arndt, 2018; Barnes et al., 2021; Barnes, 2023; Barnes et al., 2023). This collection includes whole-rock geochemistry of 5767 samples, primarily from the Archean basement of Western Australia, and has been made available through both the CSIRO repository and EarthBank platform to increase the visibility of this data and further research using this vital resource (Barnes and Devaraju, 2024). Here we demonstrate the application of this dataset as a template for evaluation of mineralisation targets using the functionalities provided within EarthBank.

Magmatic mafic-ultramafic intrusion hosted Ni-Cu-Co deposits

constitute ~37 % of global Ni–Co endowment (Mudd and Jowitt, 2014), and are of vital importance for transition to green technologies. Deposits form when a parent magma passes through crustal conduit system where it assimilates sulphur (usually as sulphide) from country rocks, and this sulphide melt then reacts with parent magma to sequester chalcophile elements such as Ni, Cu, Co and PGEs. Deposits then require a physical process to focus and separate sulphide liquid to concentrate chalcophile elements to extractable levels (e.g. Keays, 1995; Barnes and Lightfoot, 2005; Naldrett, 2011; Lightfoot et al., 2012; Barnes et al., 2016; Barnes, 2023).

Magmatic Ni-Cu-Co-PGE deposits are often difficult to identify during exploration as they commonly lack alteration haloes or chemical footprints which expand the observable chemical anomaly around mineralisation (Barnes, 2023). Additionally, host intrusions are often small; sometimes not much larger than deposits they host, and structural controls are typically cryptic (Lightfoot et al., 2012; Barnes, 2023). This renders electromagnetic surveying problematic due to presence of barren conductors or deep, small deposits (Peters, 2006). Successfully targeting these deposits, therefore, requires predicting prospective terranes, host intrusions and deposition sites using a combination of tools coupled with sound understanding of the formation mechanisms and signatures of such mineralisation.

Using the Barnes komatiite Ni-Cu-Co-PGE geochemistry collection and data interrogation tools within EarthBank, some prospective areas for this mineralisation style may be identified to demonstrate the application of the platform for exploration. Firstly, a high-level overview of the regional geology can be performed for the sampled regions in Western Australia (Fig. 5), which shows the Archean and Palaeoproterozoic basement terranes of the Yilgarn Craton and Pilbara Craton host the majority of samples within this compilation. These terranes are of suitable age and lithology to provide prospective host terranes for

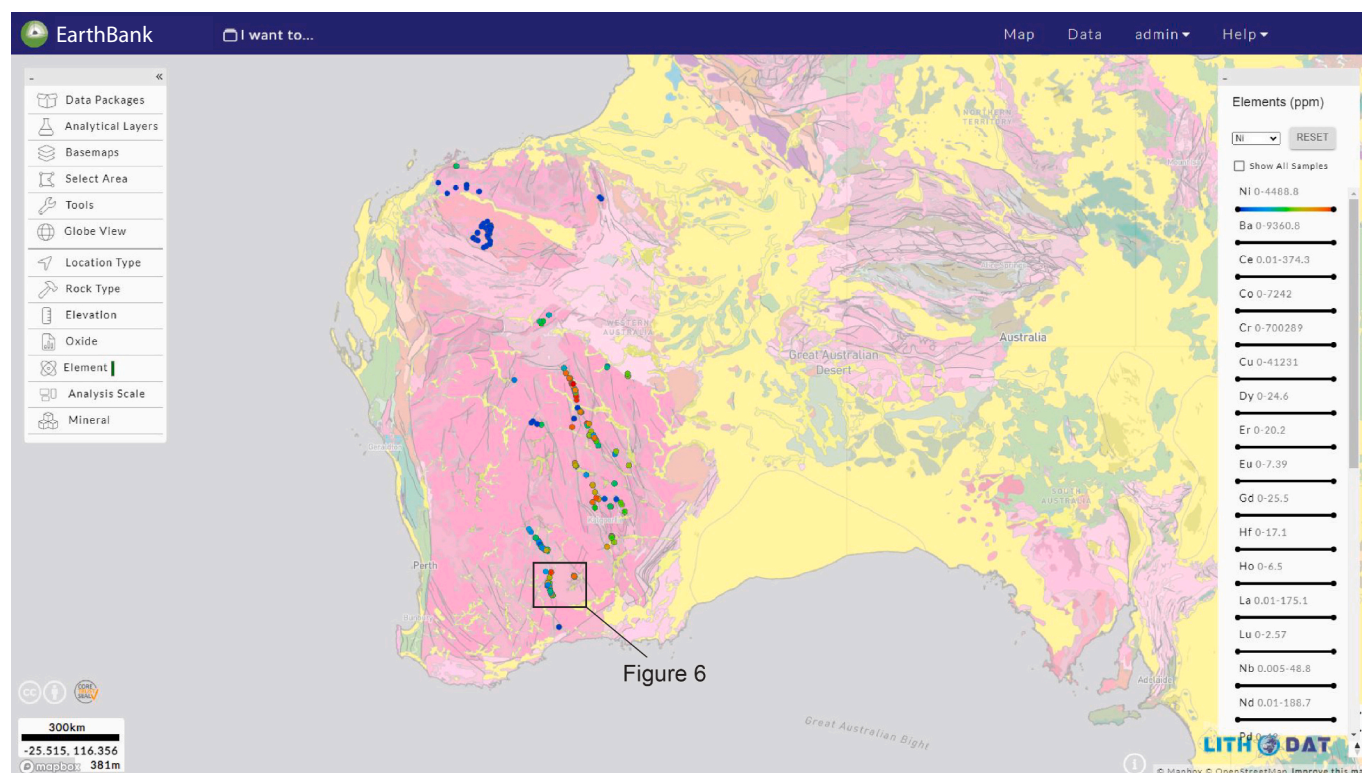


Fig. 5. Geological map of Western Australia, with locations of the Barnes Ni-PGE geochemistry collection overlain in EarthBank. Geological basemap is sourced from Macrostrat (Peters et al., 2018) and can be queried in the online platform to identify basic age and lithology information of geological terranes and strata.

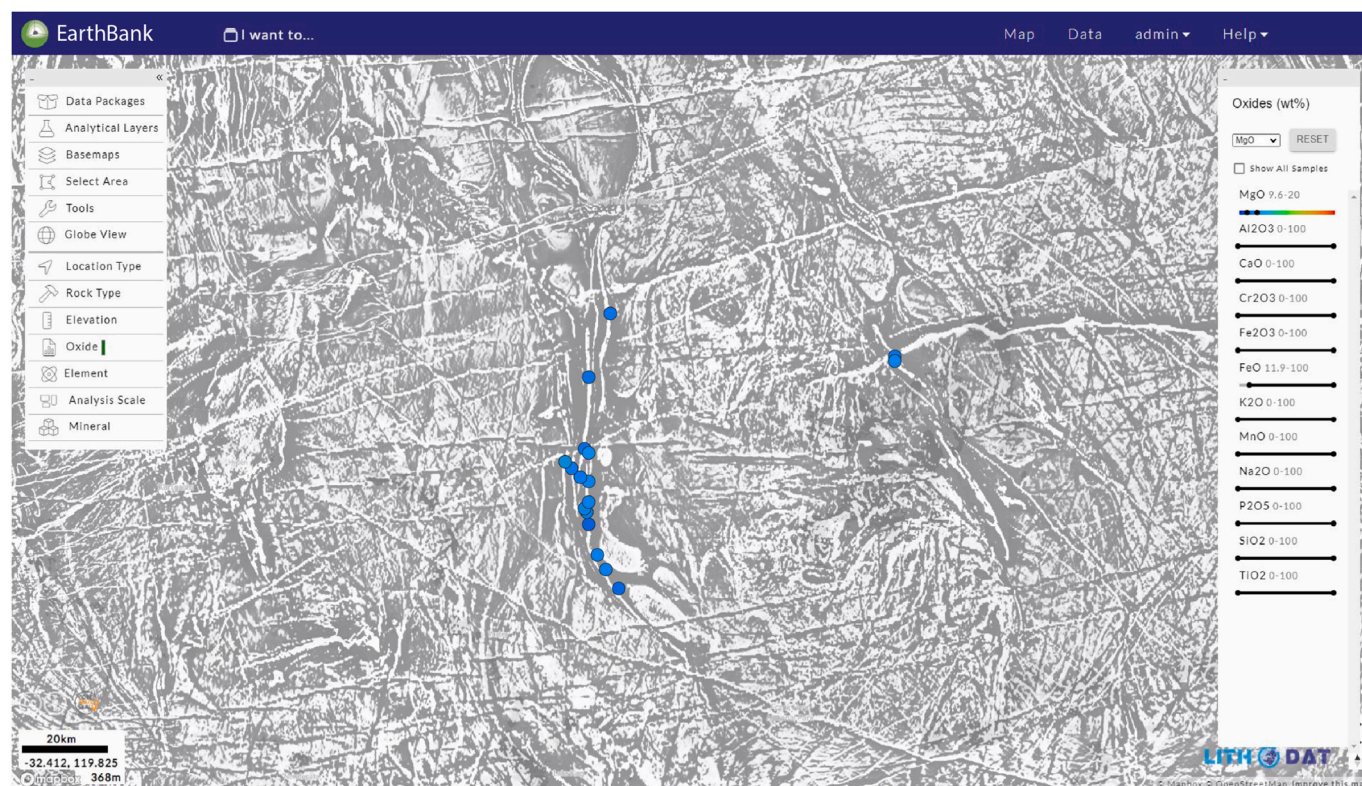


Fig. 6. Magnetics map of the southern Yilgarn Craton with Barnes Ni-PGE geochemistry samples overlain in EarthBank. Location is as shown in Fig. 5, with magnetic basemap is sourced from Geoscience Australia. Whole-rock geochemistry has been filtered for samples which exhibit high FeO (>12 wt%) and medium to high MgO content (10–20 wt%).

magmatic mafic-ultramafic intrusion hosted Ni-Cu-Co-PGE mineralisation (e.g. Barnes and Lightfoot, 2005; Naldrett, 2011; Barnes et al., 2015; Barnes, 2023). Importantly, a magnetics map can summarily be used to identify magma conduit structures within these terranes, such as those observed in the southern Yilgarn Craton (Fig. 6), which are required to extract parental magma and focus chalcophile elements.

Having established a region of interest using geological and magnetic imagery, geochemical data may now be considered in detail. Magmatic sulphide ores vary globally but are commonly associated with high pressure and temperature melts, with fertile mafic-ultramafic melts enriched in FeO (> 12 wt%) and moderately enriched in MgO (10–20 wt %; Barnes, 2023). Suitable samples may be easily screened using the EarthBank platform to identify areas for further investigation (Fig. 6). A further distal footprint of Ni sulphide deposits is the presence of cumulate rocks with evidence of silicate and sulphide extraction (Barnes, 2023), which may be detected chemically by use of the Cu/Zr ratio. Chalcophile elements such as Cu will partition to the sulphide liquid phase, while lithophile elements such as Zr will remain in the silicate melt phase. Cumulate derived rocks with a Cu/Zr ratio above that of the primitive mantle (~5) are indicative of magmas that have experienced accumulation of sulphide liquid and are prospective for magmatic Ni-Cu-Co-PGE mineralisation (Barnes, 2023). The Cu/Zr ratios of many of the mafic-ultramafic samples from the southern Yilgarn Craton are above that of the primitive mantle (Fig. 7), which coupled with magnetic and lithological indicators marks this as a prospective region for magmatic Ni-Cu-Co-PGE mineralisation. This conclusion based on exploration within EarthBank is supported by the presence of a number of Ni mines within the area, such as the Cosmic Boy, Spotted Quoll, and Flying Fox mines. This example helps demonstrate how the flexible tools and functions within EarthBank may be applied to any region in the world to conduct regional to camp scale mineral exploration using the growing wealth of geochemical data.

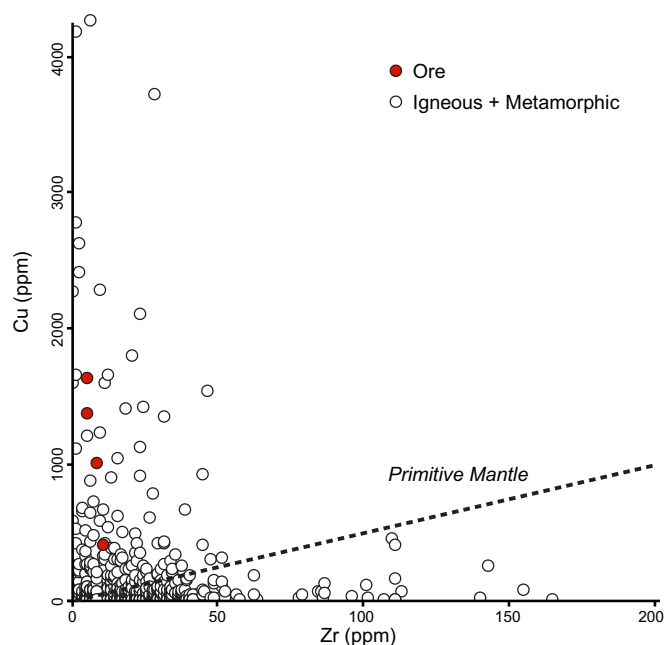


Fig. 7. Cu vs. Zr plot of mafic-ultramafic whole-rock samples from the Barnes Ni-PGE geochemistry collection in the southern Yilgarn Craton. Samples included are as shown in in Figs. 5–6. Materials which have been classified as ‘Ore’ and ‘Igneous’ or ‘Metamorphic’ in the collection were grouped and coloured in the EarthBank platform, and the plot was exported as a .pdf file for post processing. Figure editing involved adding a key to identify sample classifications and the addition of a dashed line to denote the primitive mantle Cu/Zr ratio of ~5, to demonstrate figure creation using platform outputs with minimal processing.

3.3. Trace elements – Ontario stream sampling

In addition to more traditional whole-rock or mineral geochemical sampling reflective of the properties and character of a geological feature at a single spatial location, geochemical analysis may also be applied to mixed sediment and detritus, such as that found in fluvial or marine systems (e.g., Carranza, 2004; Singh, 2010; Armstrong-Altrin et al., 2015; Kirkwood et al., 2016; Amorosi et al., 2022; Wang et al., 2022). These mixed component analyses can be used to infer the characteristics and/or relative detritus and geochemical contributions of the wider catchment region, by sampling from representative sediments within a fluvial, marine or lacustrine system.

Here we present an extensive dataset of stream sediment geochemistry from southern Ontario, Canada, released by the Ontario Geological Survey (Dyer and Burke, 2012). This dataset features 2113 samples analysed for 59 trace, minor and major elements, and demonstrates the potential for resolving lithological, geomorphological, geochemical influences on drainage samples using a relational data structure tied to interactive web visualisation. These analyses do not constitute the complete elemental budget of drainage sediment samples, rather may be considered as indications of enrichment or depletion in key elemental abundances relative to surrounding regional stream sediments. All data presented here is openly available at <https://doi.org/10.57744/LDLDEB878DE5> (Noble et al., 2024).

The study area comprises three major geological terranes, the oldest of which is Precambrian basement of the Frontenac Terrane in the Grenville Supergroup which is predominantly comprised of metamorphic rocks (e.g. Rivers et al., 1989; Marcantonio et al., 1990; Easton and Stott, 1993; Davidson, 2008; Chiarenzelli et al., 2015). In the NE sample area, basement rocks are unconformably overlain by Cambrian–Ordovician sedimentary rocks of the Ottawa Embayment (e.g. Sharma et al., 2003; Dix and Jolicoeur, 2011; Lowe et al., 2017; Lowe et al., 2018; Oruche et al., 2018; Oruche and Dix, 2021), whereas samples in the W–SW were collected from drainages cutting Ordovician–Devonian sedimentary rocks of the Michigan Basin and Appalachian Basin (Howell and van der Pluijm, 1990, 1999; Ettensohn and Brett, 2002; Armstrong and Dodge, 2007). The Michigan and Appalachian basins are broadly contiguous within southern Ontario, separated by structural highs of the NE striking Findlay Arch and Algonquin Arch (Ettensohn et al., 1994).

At a regional scale, the strongest spatial geochemical trends are consistent with the composition of the underlying bedrock, and major elements such as Ca and P demonstrate strong correlation with the lithology in which stream sediments were sampled. Ca is significantly elevated in Palaeozoic basin regions relative to pre-Cambrian basement localities (Fig. 8A), and is a proxy for detritus sourced from carbonates. Conversely, P is comparatively depleted in basin regions as compared to basement areas (Fig. 8B), as this is a proxy for phosphate minerals such as apatite and monazite which are rich in metamorphic rocks relative to sedimentary carbonates.

Sedimentary rocks of the Michigan Basin, Appalachian Basin and Ottawa Embayment comprise predominately of Palaeozoic limestone and dolomite, with relatively minor shale, siltstone and sandstone units (Armstrong and Dodge, 2007; Lowe et al., 2017; Oruche and Dix, 2021). The observed correlation between stream sediment samples which exhibit elevated Ca content in regions or abundant carbonate rocks (Fig. 8A) suggests a significant portion of sediment budget is sourced from local sources, causing an elevation in Ca concentration. The inverse is also observed for P (Fig. 8B), where the lower proportions of detritus sourced from Precambrian basement are present in stream sediments in basin terranes is likely reflective of comparatively lower P concentrations in basin stratigraphy than in metamorphic basement rocks of the Frontenac Terrane (Peck et al., 2013).

Divergence trends in Ca and P concentrations is, however, notably different between the Michigan and Appalachian basins to the SW, and the Ottawa Embayment in the NW (Fig. 8). Concentrations are most

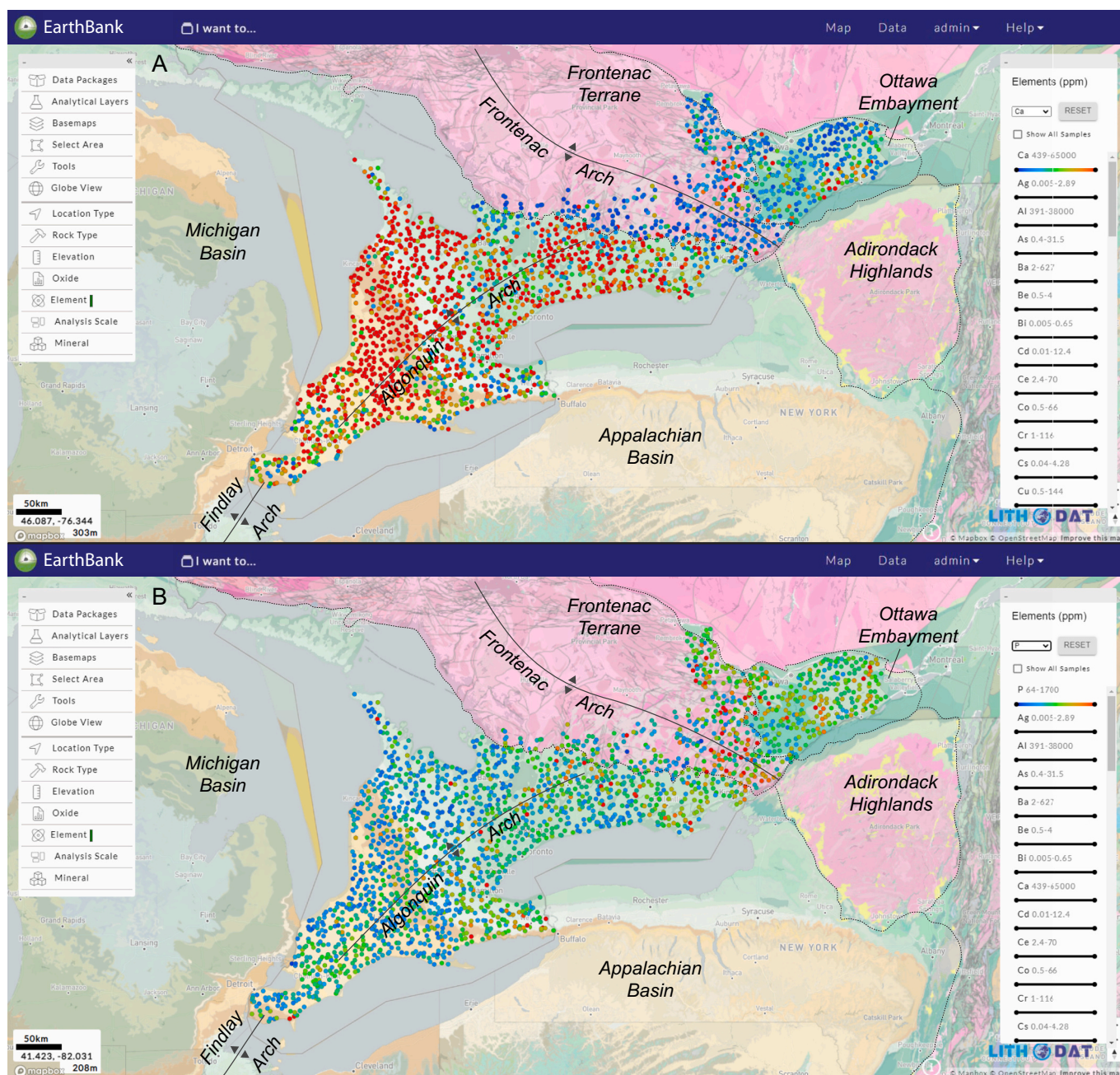


Fig. 8. Southern Ontario stream sediment geochemistry visualised within the EarthBank platform. Geological map is sourced from Macrostrat (Peters et al., 2018) and can be queried in the online platform to identify basic age and lithology information of geological terranes and strata. Measured stream sediment concentrations of Calcium (Ca; upper panel) and Phosphorous (P; lower panel) have been plotted using the platform. Major regional geological features have been annotated over the EarthBank view to provide regional context.

significantly offset from basement region samples in the Michigan Basin and Appalachian Basin, while elemental abundances in stream sediments from the Ottawa Embayment share a greater similarity with those of the surrounding Precambrian basement (Armstrong and Dodge, 2007; Lowe et al., 2017; Oruche and Dix, 2021). Additionally for trace elements, rare Earth elements (REEs) – particularly light rare Earth elements (LREEs) and middle rare Earth elements (MREEs) – are enriched in stream sediments in the Ottawa Embayment and Grenville Supergroup, while comparatively depleted in stream samples in the SW basins (Fig. 9). It is notable that the Ottawa Embayment is a comparatively small basin surrounded by elevated topography of Precambrian basement terranes at the present day, while the larger Michigan and Appalachian basins are largely surrounded by regionally extensive lakes

(Fig. 9). These trends suggest that streams draining to the Ottawa Embayment and additional glacial transport have sourced significant meta-igneous material from the surrounding basement highlands, including REE bearing phosphates such as apatite and monazite (Pan and Fleet, 2002; Spear and Pyle, 2002), while modern fluvial systems in the Michigan and Appalachian basins are largely recycling detritus from the underlying carbonaceous basins themselves.

A notable Zn-Cd-Pb geochemical anomaly is present in multiple stream sediment samples in a semi-rural region west of Lake Ontario and bounded by population centres of Guelph, Hamilton and Milton. Here drainage sediments contain Zn (≥ 1500 ppm), Cd (up to 12.4 ppm) and Pb (up to 127 ppm) concentrations significantly above both regional background levels and average crustal abundances (Fig. 10A–C; e.g.

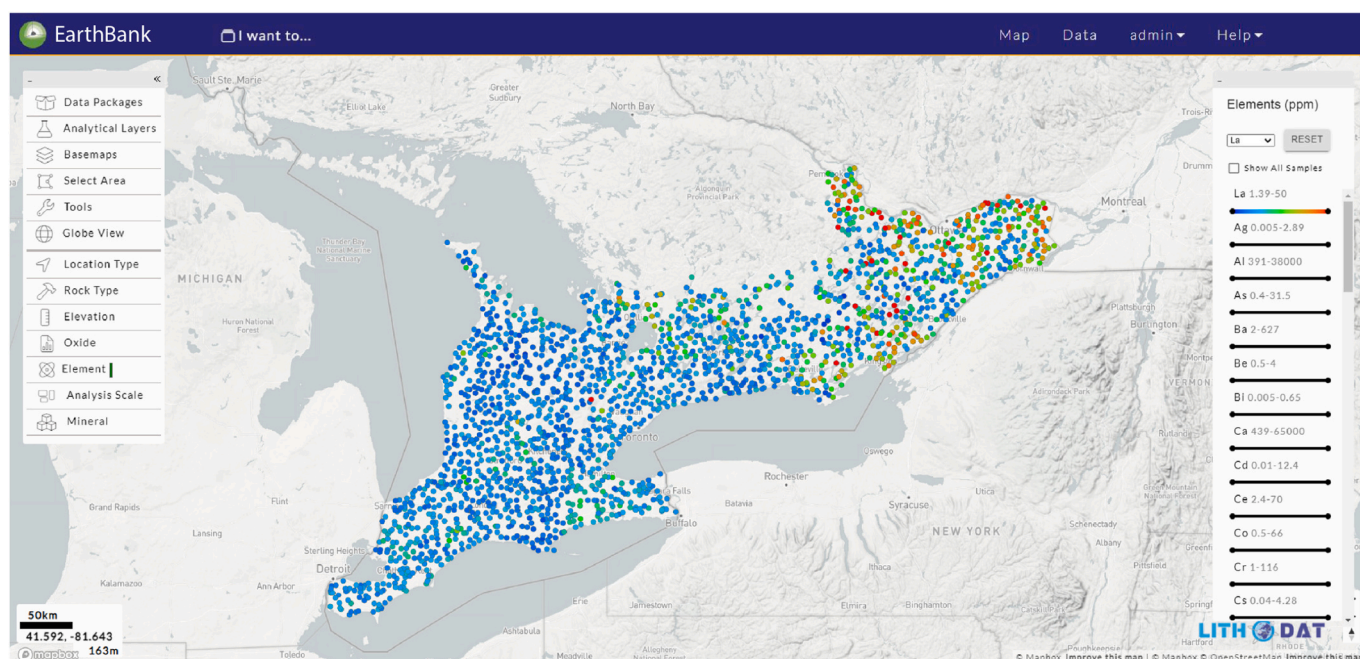


Fig. 9. Measured concentrations of Lanthanum (La) from southern Ontario stream sediment geochemistry visualised within the EarthBank platform. Geochemical data are overlaid on topographic map available through EarthBank.

Taylor, 1964; Taylor et al., 1981; Wedepohl, 1995; McLennan, 2001; Yaroshevsky, 2006; Hu and Gao, 2008). The co-occurring enrichment of these elements in soils and waterways is characteristically associated with the weathering, extraction or processing of rocks containing Pb–Zn mineralisation, specifically the ore minerals galena (PbS) and sphalerite (ZnS) (Nriagu and Pacyna, 1988; Fuge et al., 1989; Jung and Thornton, 1996; Hu and Gao, 2008; Jung, 2008).

Inspection of the satellite map layer available within the EarthBank platform reveals that elevated concentrations of Pb, Zn and Cd coincide with several quarry sites which mine limestone and dolostone for concrete aggregate, road stone, and as a metallurgical flux for the steel industry – including Dufferin Aggregates Quarry (west of Milton; 43°32′29″ N, 79°57′43″ W), Dufferin Aggregates Flamboro and Lafarge Dundas Quarries (NW of Hamilton; 43°17′34″ N, 79°59′28″ W) and Dufferin Aggregates Aberfoyle Quarry (south of Guelph; 43°27′48″ N, 80°08′43″ W; Fig. 10D). The bedrock formations being mined in these localities include the Guelph and Eramosa Formations which are fossiliferous saccharoidal dolostone of Silurian age (Brunton and Brintnell, 2011). The Lafarge Dundas Quarry is known for Mississippi Valley Type Zn–Pb mineralisation with sulphide minerals commonly found in vugs, fractures and porous reefal bioherms (Kwong, 1973). Trace element analysis of five regional samples of Eramosa Formation were reported by Rowell (2015; Appendix E) to have anomalous concentrations of Zn (8–7707 ppm; average = 1609 ppm), Cd (0.01–4 ppm; average = 1.1 ppm) and Pb (2–17 ppm; average = 8.5 ppm), which are below observed drainage sediment anomalies of Cd and Pb. This suggests that if this mineralisation is the source of metals in the local drainages, then there must be some form of physical concentration mechanism which is occurring within the rivers and streams such as density sorting of heavy mineral layers such as sulphides. Alternatively, the anomalous metals in stream sediments may be sourced from more highly mineralised regions within the Guelph and Eramosa Formations than those reported by (Rowell, 2015).

4. Discussion

As geochemical data acquisition capabilities continue to expand, and data interrogation and synthesis tools increase in both popularity and

complexity, the need to bridge the gap between geochemical data producers and end (re)users becomes paramount. EarthBank has been designed to address multiple facets of responsible and innovative data service to the geochemical community, taking the unique approach of covering a broad span of the data lifecycle from generation to interpretation, publication, and community reuse. As the platform provides the option for users to mint Digital Object Identifiers (DOIs) for these data resources at the data package level (Fig. 1A), this data is not only findable but importantly is citeable. This provides an avenue for publishing research data in accordance with the open data practices of many research journals, removing the burden on scientists to re-upload their data to a secondary platform by instead providing this service in a platform tailored to serve as a tool across the research workflow. This model has already been adopted to examine and publish datasets for peer-reviewed studies elsewhere, including a global database of kimberlite geochemistry (Giuliani et al., 2024), multi-method investigation of thermal evolution within the East African rift system (McMillan et al., 2024), and provenance tracing of sediments within the central Australian Cooper-Eromanga Basin (Nixon et al., 2024).

Data and metadata held within EarthBank are standardised to formats following international best practice recommendations (e.g. Wilkinson et al., 2016; Chamberlain et al., 2021; Klöcking et al., 2023) and working group recommendations. The standardised (meta)data allows for combined plotting and interrogation of datasets from multiple sources within EarthBank, as shown in the case studies above, but more importantly provides the framework for data sharing and interoperability with other repository services. The vocabularies associated with the EarthBank are openly available through the ARDC hosted RVA service at <https://vocab.ardc.edu.au/viewById/689>, and available with the model schema tables in Tables S1–S9 in Appendix A.

Unlike many data services, EarthBank permits – and indeed encourages – geochemical data and metadata to be uploaded immediately following collection and prior to publication. The advantages of this approach are twofold; firstly, collecting data directly from the producing laboratories or analysts provides a greater likelihood that rich and complete metadata will be captured, as opposed to merely that which is provided upon publication. Secondly, uploading results following acquisition provides a pathway for capturing results which do not make

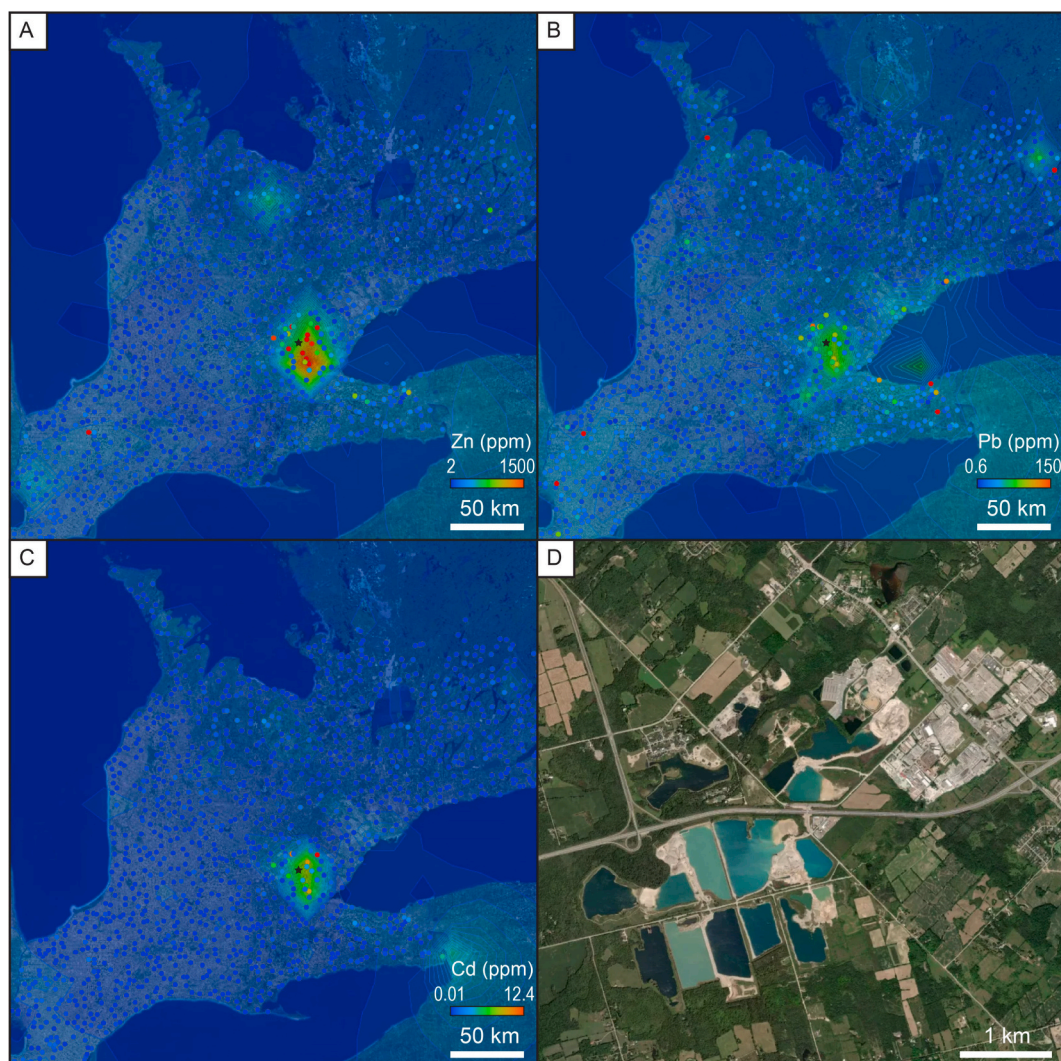


Fig. 10. Interpolation maps of southern Ontario stream sediment geochemistry overlaid on satellite imagery maps within the EarthBank platform, which demonstrate use of web-based interpolation to identify geochemical distribution. (A) Inverse distance weighting interpolation map of Lead (Pb); (B) inverse distance weighting interpolation map of Zinc (Zn); (C) inverse distance weighting interpolation map of Cadmium (Cd); (D) satellite image of pit workings and tailings ponds near the town of Aberfoyle at the centre of geochemical anomaly observed in stream sediment concentrations of Pb, Zn and Cd, location as shown by black star in panels A–C.

it to traditional publications due to factors such as project direction, time required to compose and revise manuscripts, or perceived lack of novelty in the data. EarthBank provides an alternate route of publishing as a data resource (with DOIs) for results which remain scientifically valid yet otherwise would not have been available to the community. This allows for such data to be used by others to advance their own research, but also allows the producing researcher(s) to be cited for data contribution to acknowledge the source of data. Throughout the pre-publication stage, both instances are supported by privacy control settings as detailed in [Boone et al. \(2022\)](#) to maintain the advantages of early data capture, while ensuring materials remain confidential to the authors until ready for publication.

In addition to providing the spatial visualisation and plotting functionalities as detailed in the above case studies, EarthBank also provides a more basic, yet crucial, service as a repository of the data and metadata associated with the research. Once data is made public, with a DOI assigned to the dataset if desired, this allows others to reuse information following publication, and provides the resources for appropriate attribution and citation of this data. These combined resources for data management are designed to make geochemical research data findable, accessible, interoperable and reusable (FAIR; [Wilkinson et al., 2016](#)),

while offering a service beneficial to the active research process.

5. Conclusions

Major, minor and trace element geochemistry abundance is widely used across a range of disciplines and applications within Earth science, as probably the most ubiquitous data type collected by research, government and commercial laboratories. Due to the many applications and producing laboratories, this information commonly resides in siloed repositories or inaccessible storage systems within unaligned data formats. These hurdles dampen the value of otherwise viable data resources, by reducing the ease or possibility of access by the community. To help address this inefficiency, the AuScope Geochemistry Network has developed an openly accessible inorganic geochemistry schema to store, explore and share data within the specially designed EarthBank platform. This tool allows researchers to upload and interpret their own data or supplement with publicly available resources, helping to extend the scientific longevity of data which may otherwise be lost. As demonstrated through a number of diverse case studies, this resource is flexible enough to accommodate geochemical data pertaining to a wide range of Earth science applications, including igneous systems,

mineralisation and exploration, and detrital provenance studies. By providing this resource for the community, EarthBank offers a tool for geoscientists both to advance their own research, but also to preserve and extend the results of their studies to be reused and enhanced by scientists across the globe, providing a FAIR data environment to progress the pursuit of Earth science for all.

Data and materials availability

All data presented in this article are available through public databases as follows: the Sunda-Banda Arc dataset is available at <https://doi.org/10.58024/AGUAEBE2CBB4>; the Ni-Cu-Co-PGE Mafic-Ultramafic Mineralisation dataset is available at <https://doi.org/10.25919/p5v1-j775>; and the Ontario Stream Sampling dataset is available at <https://doi.org/10.57744/LDLDEB878DE5>. Vocabulary resources associated with this model are openly available as machine readable formats via the Research Vocabularies Australia (RVA) service at <https://vocabs.ardc.edu.au/viewById/689> and associated vocabularies, or in static tables in Appendix A.

CRedit authorship contribution statement

Angus L. Nixon: Writing – original draft, Visualization, Methodology, Investigation, Data curation, Conceptualization. **Samuel C. Boone:** Writing – review & editing, Visualization, Methodology, Conceptualization. **Yoann Gréau:** Writing – review & editing, Investigation. **Fabian Kohlmann:** Writing – review & editing, Visualization, Data curation. **Moritz Theile:** Methodology, Data curation. **Wayne Noble:** Methodology, Data curation. **Alexander Prent:** Supervision, Conceptualization. **John Foden:** Data curation. **Steven Barnes:** Data curation. **Ashlea N. Wainwright:** Investigation. **Benjamin Wade:** Writing – review & editing, Investigation. **Karsten Goemann:** Writing – review & editing, Investigation. **Sarah Gilbert:** Writing – review & editing, Investigation. **Noreen J. Evans:** Writing – review & editing, Investigation. **Hayden Dalton:** Writing – review & editing, Conceptualization. **Bryant Ware:** Writing – review & editing, Supervision. **Brent I.A. McInnes:** Writing – review & editing, Supervision, Project administration.

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.chemgeo.2025.123092>.

Data availability

All data used are available through public databases at following links: <https://doi.org/10.58024/AGUAEBE2CBB4>; <https://doi.org/10.25919/p5v1-j775>; <https://doi.org/10.57744/LDLDEB878DE5>

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